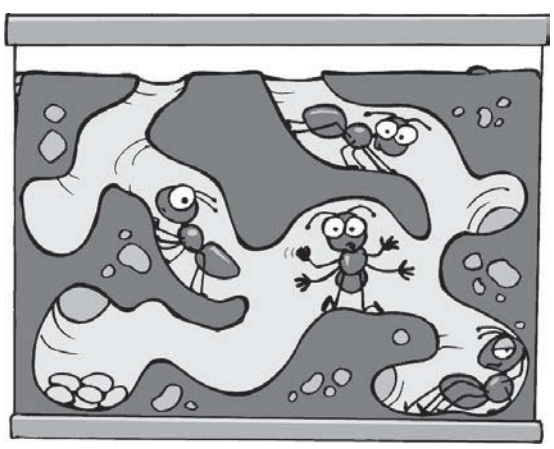
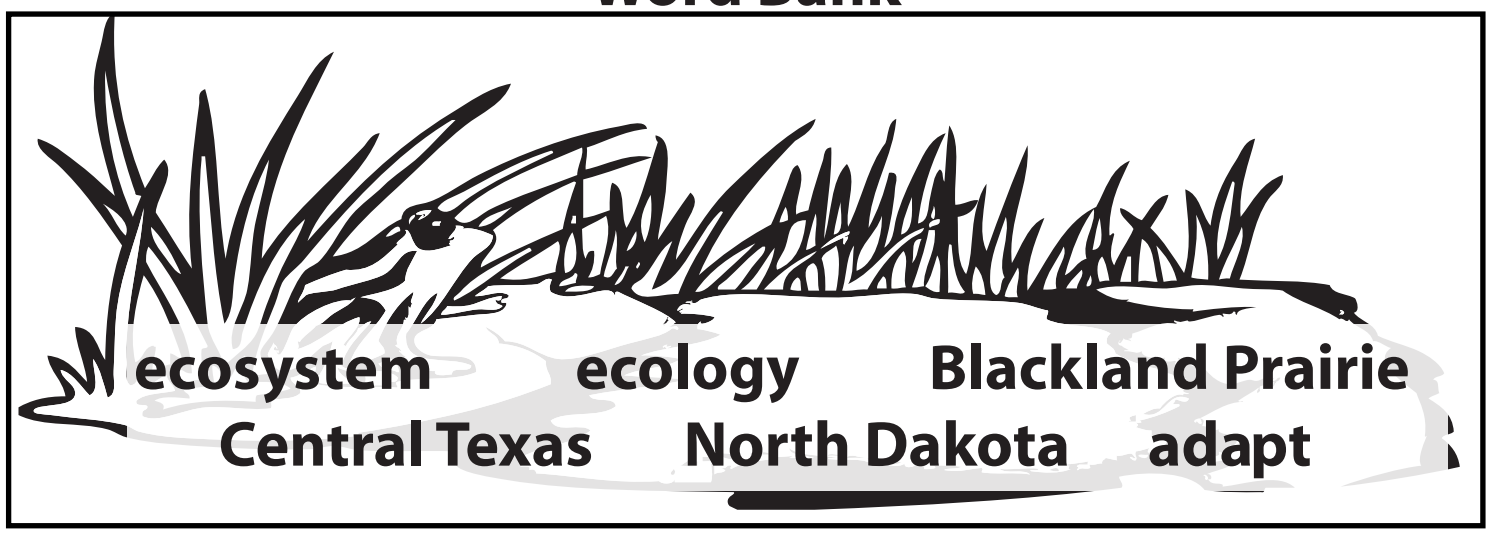
The water around you, the air above you, the rocks and the soil, and all living things make up the ecosystem.



Ecosystems can range in size from an ant farm to an entire ocean. Your ecosystem includes all of the things you need to survive. The Blackland Prairie ecosystem stretches from Central Texas to North Dakota.

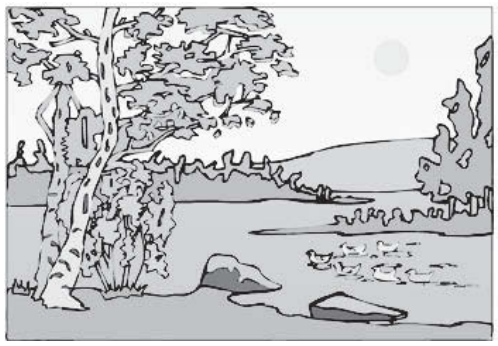
When scientists study how all of these things react to each other, it is called ecology. By studying ecology we know if any changes occur and we can adapt to those changes as needed.

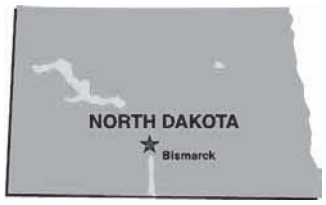
Word Bank



The water around you, the air above you, the rocks and soil, and all of the living things make up the

_____.





The land between Central Texas and North Dakota is considered the _____

_____ecosystem.

This ecosystem ranges from _____ _____

all the way up to _____ _____.

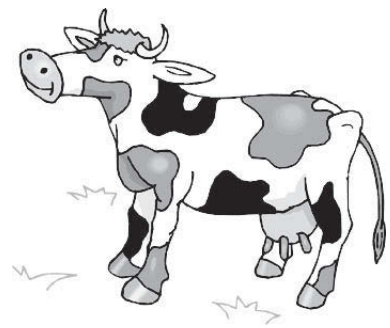
The study of all of the organisms in the ecosystem is called

_____.

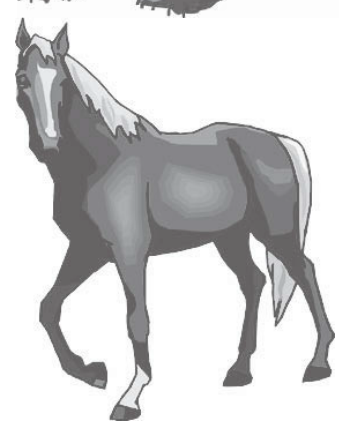
It is very important that we study ecology because it allows us to know if changes occur so we can

_____.

_____Animals are part of the ecosystem.



_____Oklahoma is part of the Blackland Prairie.



_____Ecology is the study of the weather.

_____Plants are not part of the ecosystem.

_____ It is important to study the ecosystem to make sure no drastic changes occur.

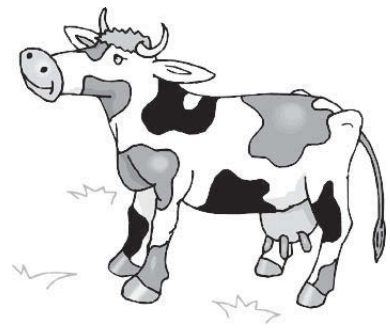
_____Ecosystems can be many different sizes.

Ecosystems- Blackland Prairie

Write true or false for each sentence.

_____Animals are part of the ecosystem.

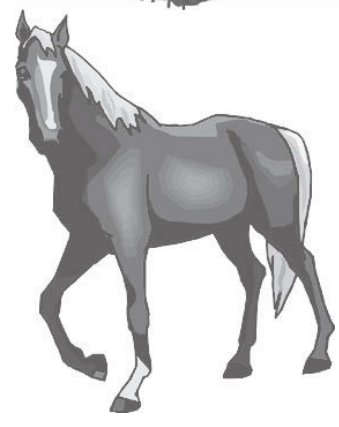
True



_____Oklahoma is part of

True

the Blackland Prairie.

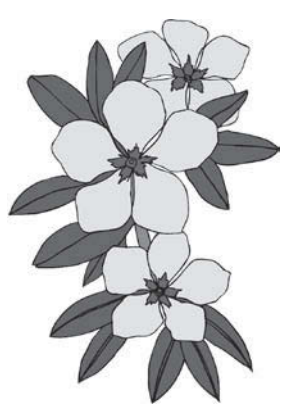


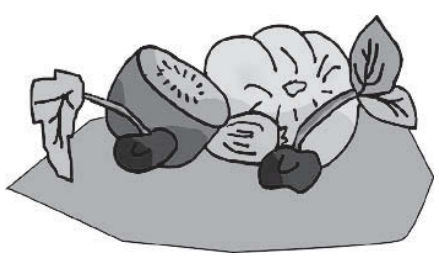
_____Ecology is the study of the weather.

False

_____Plants are not part of the ecosystem.

False





_____ It is important to study the ecosystem to make sure no drastic changes occur.

True

_____Ecosystems can be many different sizes.

True

Word Bank

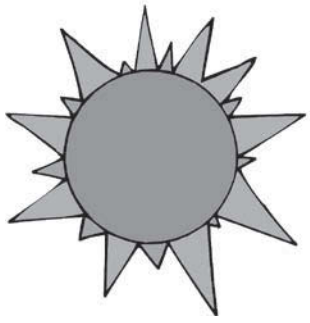
ecosystem air water Blackland Prairie Central Texas
fungus ecology organism algae rocks
animals soil North Dakota

E	A	T	H	K	B	C	H	M	A	I	R	K	L	B
C	M	A	B	N	I	K	O	I	G	D	C	B	K	L
O	S	L	R	T	O	J	F	U	N	G	U	S	F	A
S	D	G	K	H	X	R	T	T	Y	J	H	K	G	C
Y	F	A	J	C	C	Q	T	R	H	K	J	L	G	K
S	G	E	L	E	D	P	I	H	T	L	K	O	E	L
T	H	T	O	N	G	I	G	L	D	O	D	L	C	A
E	T	Y	F	T	T	W	R	J	F	A	S	E	S	N
M	V	J	E	R	A	J	A	B	D	T	K	C	H	D
K	F	H	A	A	N	H	E	T	V	R	T	O	J	P
J	S	O	I	L	I	Y	S	Q	E	E	E	L	T	R
G	E	G	X	T	M	F	C	W	D	R	W	O	P	A
T	D	F	S	E	A	B	V	A	X	F	C	G	L	I
H	F	D	G	X	L	F	F	S	S	V	Y	Y	O	R
N	G	S	H	A	S	O	R	G	A	N	I	S	M	I
R	O	C	K	S	J	C	R	C	R	D	I	K	Q	E

Ecosystems- Blackland Prairie

Find the words in the Word Search.

An ecosystem is divided into two parts, abiotic factors and biotic factors.



Abiotic factors are non-living organisms like water, minerals, sunlight, and air.

Biotic factors are living organisms such as animals, plants, and fungus. Both groups are very important to the ecosystem.

Abiotic factors, or non-living groups, are producers.

They produce oxygen and food. The right abiotic factors help make it possible for organisms to live. Without water or sunlight an ecosystem would not survive. Animals are called consumers; they eat and breathe what the non-living organisms produce and also produce carbon dioxide.

Abiotic and Biotic

Read the information and trace the dotted words.

Word Bank

abiotic	**living**	**biotic**	**non-living**
producers	**consumers**	**oxygen**	**carbon dioxide**

An ecosystem is divided into two parts, the _____ factors and the _____factors.

An abiotic factor is a _____ thing such as air, water, and soil.

A biotic factor is a _____ thing such as an animal or plant.

Non-living things are considered _____, because they produce all of the essential needs of the plants and animals.

Animals are considered _____because they consume, or eat, other plants and animals.

Abiotic organisms produce the _____that the animals need.

Animals give off _____ _____, which plants use to make food.

Abiotic and Biotic *Fill in the blanks.*

The non-living part of an ecosystem are called the

_____ _____.

Water, minerals, sunlight, and air are _____-_____

_____.

_____ _____are considered _____,

because they survive off of the _____made by non-living things.

Plants and non-living things are called _____ because

they make the _____ animals breathe.

_____ _____ are the living parts

of an ecosystem.

Abiotic and Biotic *Fill in the blanks.*

The non-living part of an ecosystem are called the

_____ _____.
abiotic **factor**

Water, minerals, sunlight, and air are _____-_____
non-living

_____.
organisms

_____ _____are considered _____,
living **organisms** **consumers**

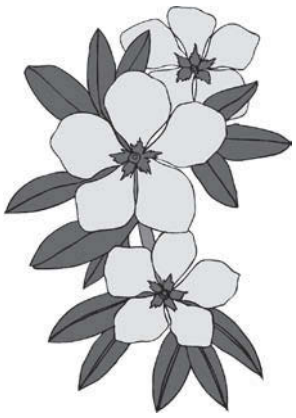
because they survive off of the _____made by
organisms
non-living things.

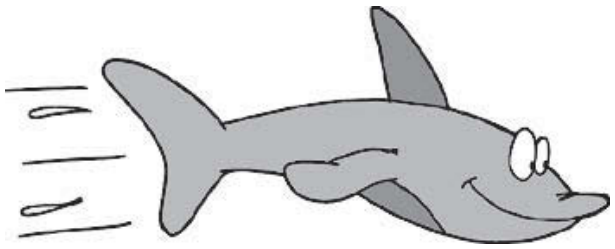
Plants and non-living things are called _____ because
producers

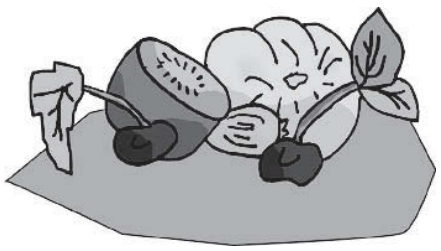
they make the _____ animals breathe.
oxygen

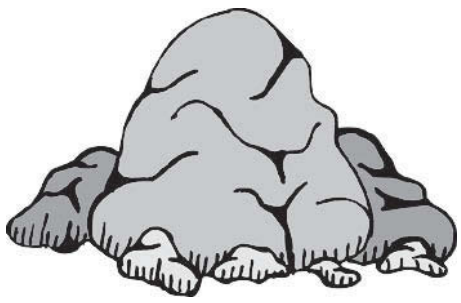
_____ _____ are the living parts
Biotic **factors**

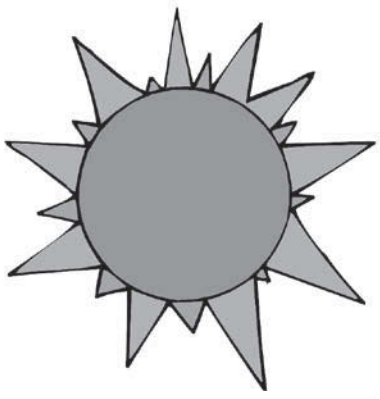
of an ecosystem.







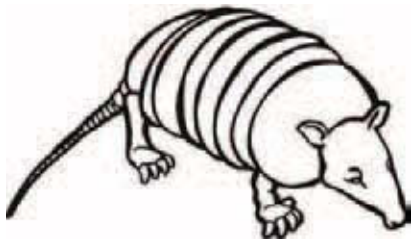




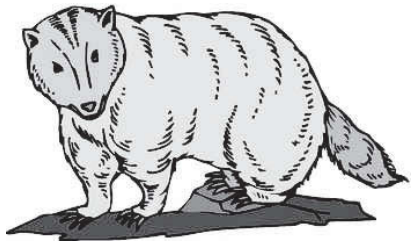
Abiotic and Biotic

Circle the biotic factors.

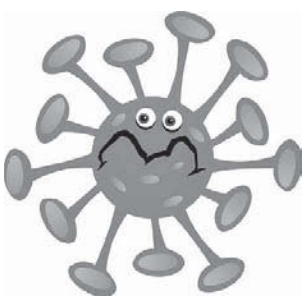
The Blackland Prairie is made up of many different organisms. Each of these organisms, whether it is an animal, plant, or bacteria, all belong to their own species or group. When all of these species live in one area, they form a population.



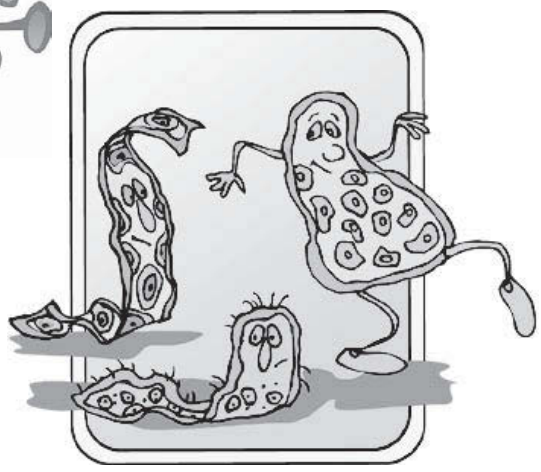
The species that live in the Blackland Prairie make up its population.



Its population includes everything from armadillos and badgers



to the smallest organisms such as algae, bacteria and fungus. All of these populations make up the ecosystem's community.

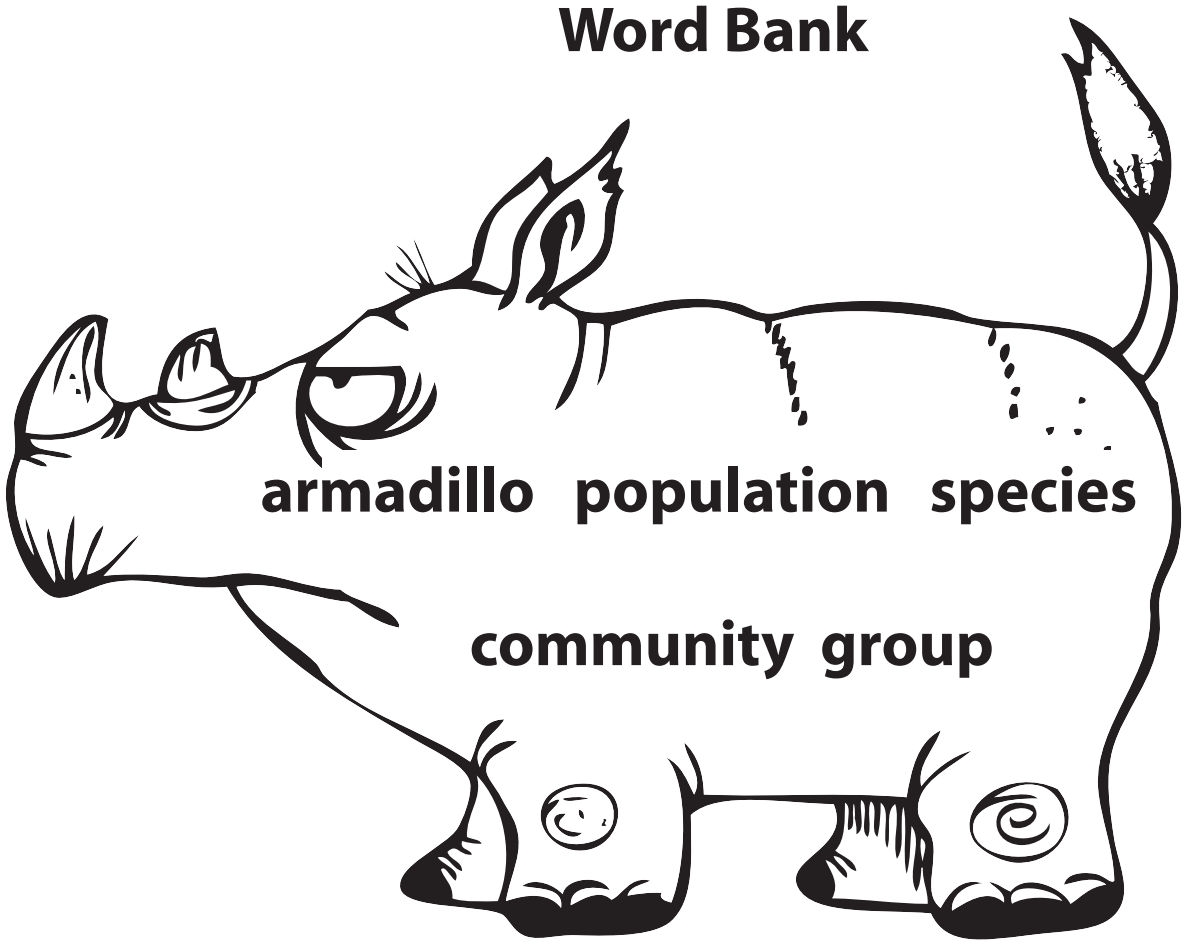


Organizing and labeling groups like this is very helpful to scientists by helping them track how each member of the ecosystem survives.

Population and Community

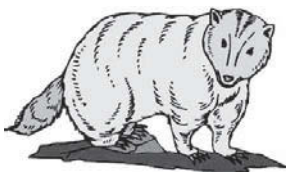
Read the information and trace the dotted words.

Word Bank



All organisms belong to groups called _____.

All of the organisms living in the same area make up the _____.



Animals such as the _____, badger, spade foot toad, and bald eagle make up the Blackland's population.

Scientists like to _____ animals together so they can track which animals may be in danger.

Population and Community *Fill in the blanks.*

What are all of the populations in an ecosystem called?

_____

Who would benefit from the grouping of animals?

_____

All organisms are members of a _____.

What are all of the organisms of a species called when they live in the same area?

_____

What is the ecosystem between Central Texas and North Dakota?

_____

What species do you belong to?

_____

What are all of the populations in an ecosystem called?

_____

community

Who would benefit from the grouping of animals?

_____

scientists

All organisms are members of a _____.

species

What are all of the organisms of a species called when they live in the same area?

_____

population

What is the ecosystem between Central Texas and North Dakota?

_____

Blackland Prairie

What species do you belong to?

_____

Human

You are on a vacation to a new ecosystem. Draw the plants and animals you see and write about it to your friend.

The place where an organism lives is called its habitat.

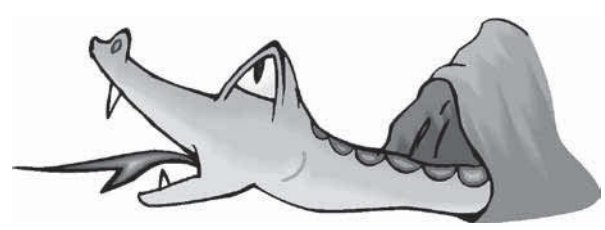
The Blackland Prairie is full of habitats.

The spade foot toad's habitat is the pond, the eagle makes nests in the trees, and the snake burrows beneath the ground.

All of the habitats belong to one ecosystem.

All of the animals have a role in the community of this ecosystem.

This role is called its niche. The niche includes what a species eats, what eats the species, what environment it needs to live in, and when the species is active.



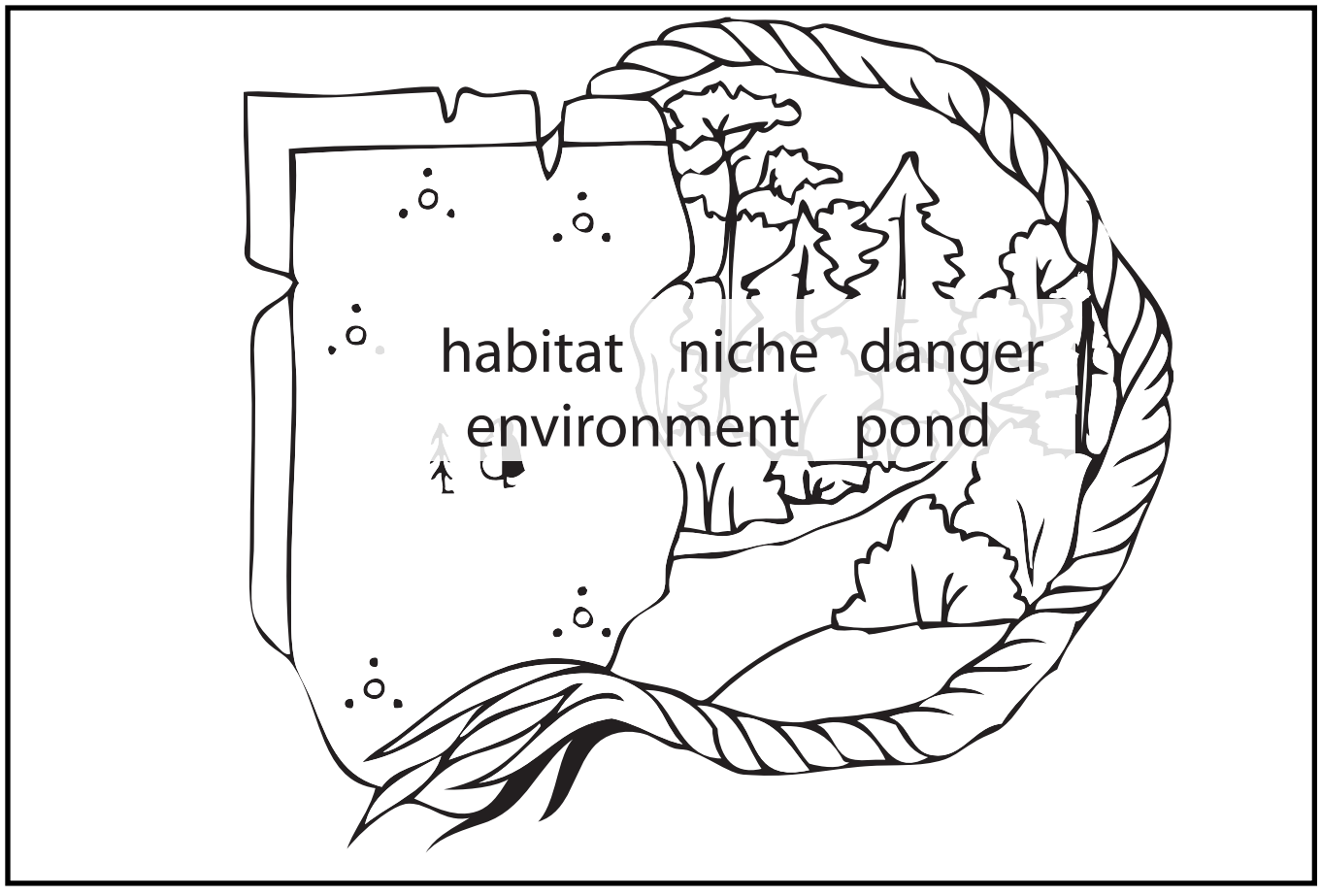
Scientists study all of these to make sure the habitat is healthy or if it may be in danger.

If one species is killed off for any reason it affects all of the animals in the ecosystem.

Habitats and Niches

Read the information and trace the dotted words.

Word Bank



The place that is home to an organism is called its

_____.

The spade foot toad's habitat is the _____.
The role of the spade foot toad in its habitat is its

_____.

The niche includes what an animal eats, what kind of

_____ it needs, and when it is active.

A scientist's job is to watch the habitat and make sure each species is safe, as well as to make sure the habitat is not in

_____.

Habitats and Niches *Fill in the blanks.*

The place where an organism lives is its _____

A. home C. habitat
B. tree D. city

The spade foot toad's habitat is the _____ .
A. nest C. woods
B. pond D. castle

The role an organism has in its community is its_____.
A. job C. career
B. population D. niche

If one animal is killed off in its habitat it affects the whole _____.

A. ecosystem C. state
B. planet D. frog family

_____study niches to make sure nothing changes in the environment.
A. Teachers C. Scientists
B. Environmentalists D. Animals

The place where an organism lives is its _____

habitat

A. home C. habitat
B. tree D. city

The spade foot toad's habitat is the _____ .

pond

A. nest C. woods
B. pond D. castle

The role an organism has in its community is its_____.

niche

A. job C. career
B. population D. niche

If one animal is killed off in its habitat it affects the whole _____.

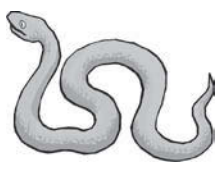
ecosystem

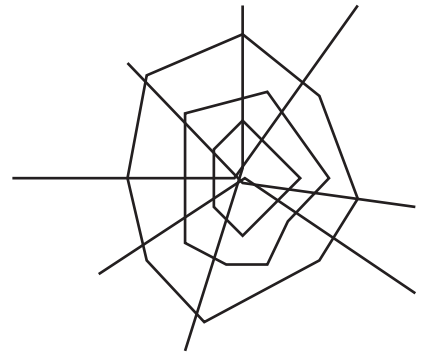
A. ecosystem C. state
B. planet D. frog family

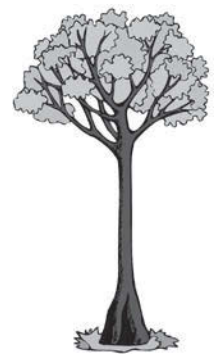
_____study niches to make sure nothing changes in the environment.

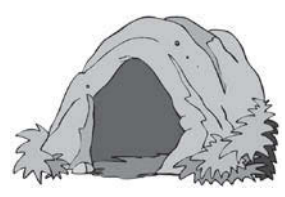
Scientists

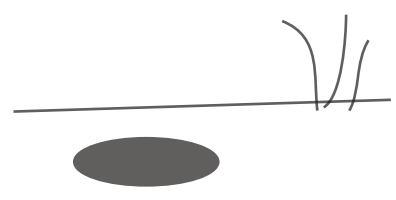
A. Teachers C. Scientists
B. Environmentalists D. Animals







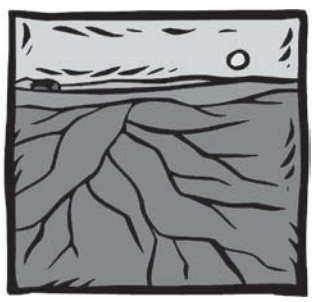


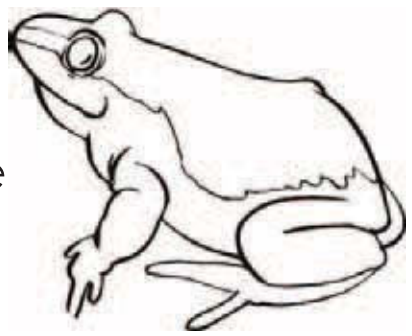


Habitats and Niches

Match the animal to its habitat.

Adaptation is a very important trait for organisms in an ecosystem. Adaptation helps animals survive changes in temperature, weather, and environment. It is an inherited trait, not a learned behavior. Every ecosystem goes through changes. One year there could be heavy rains and then the next year there could be drought. All organisms need water, so they have to adapt to the change. The eastern spaded foot toad lives in the Blackland Prairie. This animal needs water for everyday life. The toad's back feet are shaped like spades, or small shovels. They have adapted through time for digging. If water is scarce, he digs holes in different ways. Some animals are able to run fast to escape their predators. Others have sharp claws or teeth to hunt.



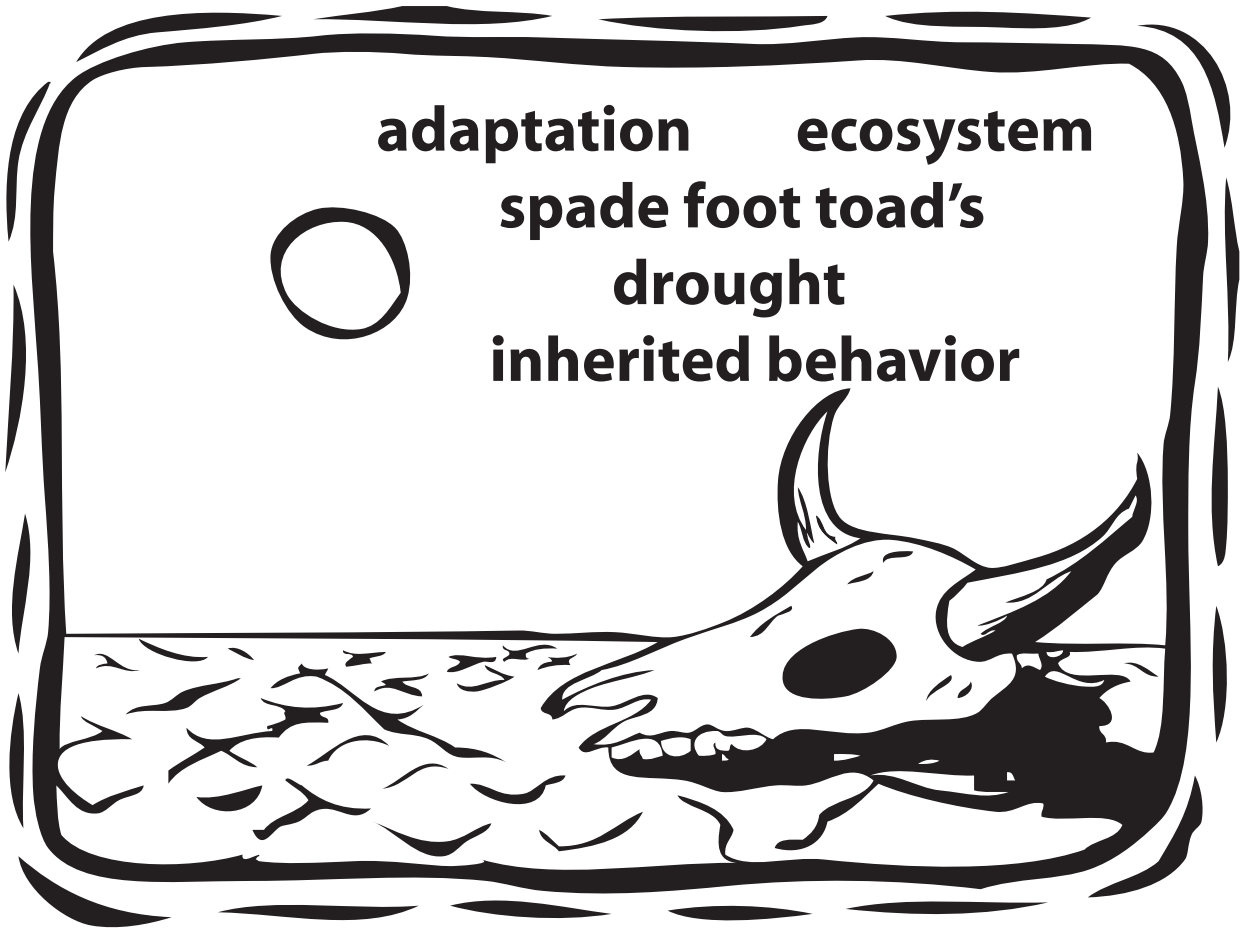


Plants rely on adaptations such as color change, aroma, or poison. All of these traits help organisms survive.

Surviving Change

Read the information and trace the dotted words.

Word Bank



_____ helps animals survive in changing environments.

Every _____ goes through changes throughout time.

One year the ecosystem could have heavy rains and then the next year there could be _____.

The _____ _____'s back legs have adapted to help it dig into the ground during a drought.

Adaptation is essential for animals' survival and is considered an _____.

Surviving Change *Fill in the blanks.*

_____ Adaptation is a learned behavior.

_____ A habitat that is good for an organism one year could be dangerous the next.

_____ The eastern spade foot toad digs holes with its nose.

_____ Weather changes could force an animal to adapt.

_____ The eastern spade foot toad has adapted to his environment.

_____ Plants can adapt to their environment.

_____ Inherited behaviors are behaviors organisms are born with..

Surviving Change

Write (T) or (F) for each statement.

_____ Adaptation is a learned behavior.
F

_____ A habitat that is good for an organism one year could be dangerous the next.
T

_____ The eastern spade foot toad digs holes with its nose.
F

_____ Weather changes could force an animal to adapt.
T

_____ The eastern spade foot toad has adapted to his environment.
T

_____ Plants can adapt to their environment.
T

_____ Inherited behaviors are behaviors organisms are born with..
T

Surviving Change

Draw yourself and tell how you would adapt.

Living in an ecosystem not only do organisms adapt to the environment, they also adapt to other organisms. In nature symbiosis occurs in the relationship between two organisms that last over time. This relationship can benefit one or both of the organisms. If it benefits both organisms it is called mutualism. If it only benefits one it is called

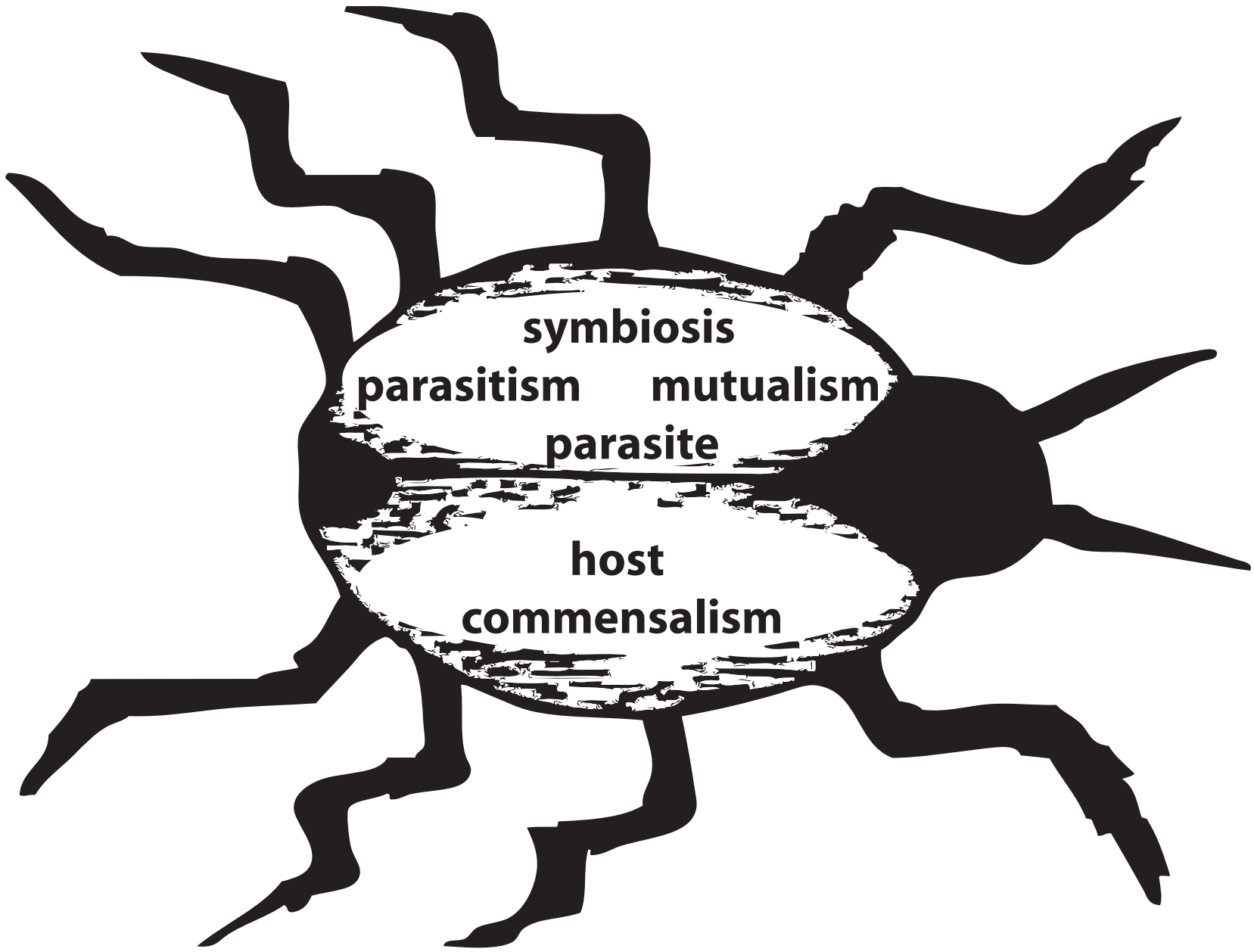
parasitism. An example of mutualism is the yucca tree and the yucca moth. The moth uses the yucca trees' pollen to wrap and protect its eggs, while doing this it is also pollinating the yucca tree's flowers. In parasitism an organism that is being used is called the host. The one using the host is called the parasite. An example of this is the dog and the flea. The flea sucks the dog's blood getting nutrients and the dog does not gain anything in return. The last type of relationship in the ecosystem is commensalism. In this relationship one organism is helped and one is not affected.

A vine that climbs a tree to get sunlight is an example of commensalism. It does not take anything from the tree but it does use it to climb up towards the sun.

Symbiosis, Mutualism, and Parasitism

Read the information and trace the dotted words.

Word Bank



In nature, _____ occurs when organisms form a relationship over time.

If a relationship benefits both organisms it is called _____.

When the relationship only benefits one organism, it is called _____.

In the relationship between a flea and a dog, the dog is considered the _____.

The _____gains all of its nutrients from the host.

If one organism is helped and the other organism is neither hurt nor helped, it is called _____

Use the Word Bank to complete the sentences.

When a vine uses a tree to gain access to the sun without taking anything from the tree it is using_____ .

The relationship between two organisms that lasts over time is called_____ .

If a relationship benefits both organisms, it is called _____ .

The relationship that only benefits one organism is _____ .

In the dog and flea relationship the dog is the _____ and the flea is the_____ .

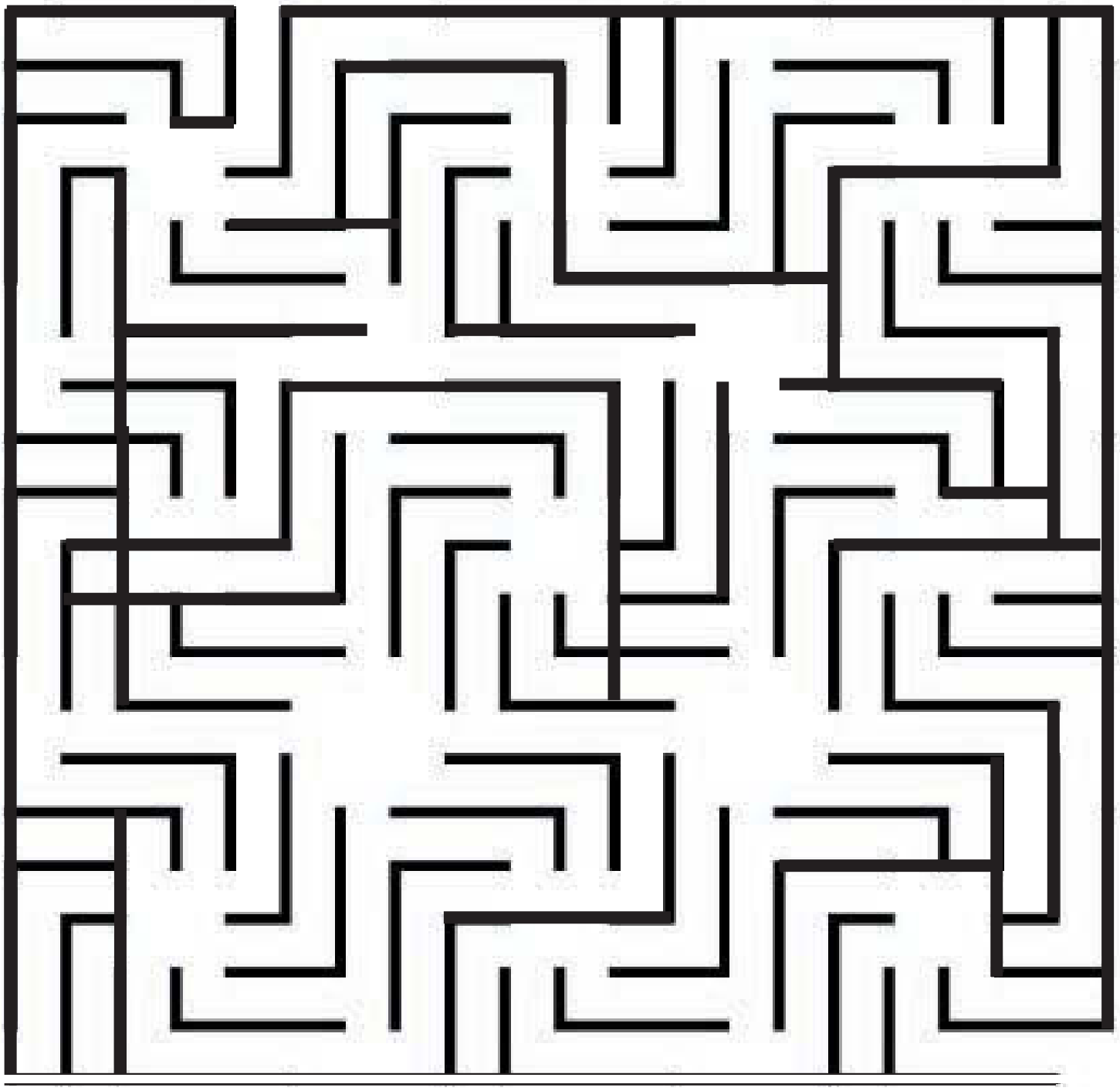
When a vine uses a tree to gain access to the sun without taking anything from the tree it is using **commensalism**.

The relationship between two organisms that lasts over time is called **symbiosis**.

If a relationship benefits both organisms, it is called **mutualism**.

The relationship that only benefits one organism is **parasitism**.

In the dog and flea relationship the dog is the **host** and the flea is the **parasite**.



Symbiosis, Mutualism, aand Parasitism

Help the parasite find the host.

Ecosystem- All the living and non-living things in an area, and their interactions with each other.

Ecology- the study of how living things and their environment interact.

Abiotic Factor- a non-living part of an ecosystem.

Biotic Factor- A living part of an ecosystem.

Population- All of the organisms of a species that live in the same area.

Community- All the populations living in the same area.

Habitat- The area in which an organism lives.

Niche- The role an organism has in its ecosystem.

Producers- Any of the plants and algae that produce oxygen and food for animals.

Consumers- Any of the animals that eat plants or eat other plant-eating animals.

Vocabulary

Review the definitions and trace the dotted words.

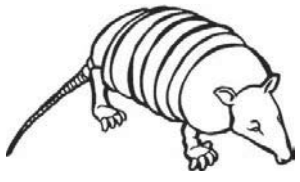
Ecosystems are all living and non-living things living in an area and their interaction with each other. Ecosystems are made up of abiotic factors

(air, water, rocks, energy) and biotic factors (plants, animals, and microorganisms).

The Blackland Prairie is the largest, most endangered ecosystem in the world. What do you need on a daily basis to

survive? We need food, water, the right temperature, and shelter. Animals need all of the same things humans do. It is important that we make wise decisions in the way we change

our environment. In trying to make progress, humans affect the ecosystem. If we change too much we may lose more than we gain.



The ecosystem is very complex because of the number of organisms that share the community. If all of nature's factors are able to do their jobs then our ecosystem will flourish.

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